COMP 4870
Intranet Planning and Development

School	School of Computing and Academic Studies
Program	Computer Systems Technology (CST) Diploma - Information Systems Option
Course Credits	7
Minimum Passing Grade	50%
Start Date	January 05, 2026
End Date	April 24, 2026
Total Hours*	105
Total Weeks	15
Hours/Weeks	7
Delivery Type	Lecture/Lab
Zero Textbook Cost	Yes
Prerequisite(s)	• 50% in COMP 3910
CRN	49852

Acknowledgement of Territories

The British Columbia Institute of Technology acknowledges that our campuses are located on the unceded traditional territories of the Coast Salish Nations of Skwxwú7mesh [1] (Squamish), səl̓ilwətaʔɬ [2] (Tsleil-Waututh), and xwməθkwəy̓əm [3] (Musqueam).

Instructor Details

Name	Medhat Elmasry
E-mail	Medhat_Elmasry@bcit.ca
Location	Online
Office Hours	Office Hours will be posted on the Learning Hub. Office hours will be via video conferencing.

Course Description

Explores Microsoft web technologies for developing and deploying e-business solutions. This course focuses on C# and ASP.NET. Two competing ASP.NET technologies, ASP.NET Web Forms & ASP.NET MVC, are covered using the latest version of Visual Studio. A term project includes implementation of a web site using either technology

Course Learning Outcomes/Competencies

Upon successful completion of this course, the student will be able to:

- Document the architecture of a well-designed ASP.NET application that has database connectivity.
- Describe the .NET Framework.
- Write computer programs using Microsoft's C# language.
- Write web pages using C# with both ASP.NET Web Forms & ASP.NET MVC.
- Use .NET HTML and Web Server controls and validation.
- Create .NET components.
- Produce and consume .NET RESTful services involving WebAPI and OData.
- Understand ASP.NET Security and authentication.
- Developing web applications with Visual Studio .NET and C#.
- Utilize .NET Debugging tools.
- Understand .NET State management.
- Implement AJAX and responsive web design in ASP.NET.
- Access SQL Server data with Entity Framework and LINQ.
- Build Single Page Applications.
- Deploy of ASP.NET web application to IIS and Azure web sites.
- Develop XML solutions involving DOM, SAX, XSLT and Schema.

Learning Resources

Material to be posted on Learning Hub

Course Goals

This course explores Microsoft's .NET technology for developing and deploying web-based solutions. It focuses on migrating database systems to an online format, using web-based tools. ASP.NET, C#, ReactJS, Entity Framework, and Visual Studio .NET. There is also a term project that will be decided later in the term.

Evaluation Criteria

Lab Work	20%	
Assignments	20%	2 assignments
Individual or group project	20%	
Mid term exam	20%	To pass the course, average of midterm and final exams must be 50% or higher.
Final Exam	20%	
TOTAL	100%	

Attendance Requirements

Regular attendance in lectures, seminars, and labs is seen as critical to student success, and may be monitored by faculty.

Unapproved absence of 2 or more classes may result in failure or forced withdrawal from the course or program.

Please see Policy 5101 - Student Regulations: https://www.bcit.ca/files/pdf/policies/5101.pdf [4]

Other information

Student Surveys (Instructional Assessments Online) - You'll complete an online survey near the end of this course to share feedback with your instructor. Surveys are anonymous, and results are released only after all final grades are submitted.

How to Access - Look for an email around Week 10 sent to your myBCIT email with a link to your survey. Click the link and choose the correct course from the "Active Tasks" list.

In-class time - Your instructor may provide 15 minutes of class time for completing the survey (individually).

Troubleshooting - If you don't receive the link or have issues, email bcit_feedback@bcit.ca before Week 11.

Tip - Make sure the survey you complete matches both your course CRN and your instructor.

Professionalism - Please provide honest, specific, and constructive feedback in your comments—your voice makes a difference.

Questions? Contact PDS Coordinator: Kirsty Barclay-Estrup**.**

Course Schedule and Assignments

Week	Starting	Essential Skill	Lab Exercise	Notes
1	Jan 5	Intro, C# & EF	OpenAI Lab	
2	Jan 12	Razor Pages & MVC	OpenAI Razor & MVC Files	
3	Jan 19	Code 1'st Development & Docker	Code 1'st	Assignment 1 posting
4	Jan 26	API, Token Authentication, & Semantic Kernel	Token Authentication	
5	Feb 2	Seed & Extend Identity + .NET Aspire	User Management	
6	Feb 9	SignalR, + Server-side & Client-side Blazor	Serverless AI	Assignment 1 due Assignment 2 posting
7	Feb 16	Serverless, Static Web Apps & React.js	React.js	Family day
8	Feb 23	Midterm Week		
9	Mar 2	TypeScript, Charts, PDF & Excel	Blazor Mix & Match	
10	Mar 9	gRPC & Localization	gRPC	Instructional Assessments Online
	Mar 16	Computing Reading Week		
11	Mar 23	TagHelpers & Cache	Tag Helper	Assignment 2 due
12	Mar 30	ML.NET	ML.NET	Good Friday
13	Apr 6	TDD & MongoDB	TDD	Easter Monday
14	Apr 13	Study/Makeup/Review/Presentation Week		
15	Apr20	Exam Week		

NOTE: Above schedule is subject to change.

BCIT Policy

The following statements are in accordance with the BCIT Policies 5101, 4501, 5103, 5104, and 7507, and their accompanying procedures. To review these policies and procedures please click on the links below.

Attendance
In case of illness or other unavoidable cause of absence, the student must communicate as soon as possible with their instructor or Program/Department Head, stating the reason for the absence. When absences result in missed safety requirements, exams, or other deadlines, an instructor or Program/Department Head may request appropriate supporting documentation, including a medical note. For all other absences, the student is responsible to ensure they seek out missed information, preferably from a classmate or recording (when available). Please see BCIT Policy 5101 - Student Regulations [5] , and accompanying procedures, for more information.

Attempts
As stated in BCIT Policy 5103 - Student Evaluation [6] , students must successfully complete a course within a maximum of three (3) attempts at the course. Students with two attempts in a single course will be allowed to repeat the course only upon permission from the Associate Dean. Students who have not successfully completed a course within three attempts will not be eligible to graduate from their respective program. For those courses or programs that have Education Council approval, the number of attempts as stated in the evaluation section of the course outline shall apply.

Academic Integrity
It is the responsibility of all students to be familiar with the Student Code of Academic Integrity. Violations of the Code, including plagiarism, cheating, misrepresentation, and academic advantage, are prohibited and will be handled in accordance with BCIT Policy 5104 – Student Code of Academic Integrity [7] , and accompanying procedures.

Accommodation
Any student who may require accommodation from BCIT because of a physical or mental disability should refer to BCIT's Policy 4501 - Accommodation for Students with Disabilities [8] , and contact BCIT's Accessibility Services (SW1 2360, 604-451-6963) at the earliest possible time. Requests for accommodation must be made to Accessibility Services, and should not be made to a course instructor or Program area.

Any student who needs special assistance in the event of a medical emergency or building evacuation (either because of a disability or for any other reason) should promptly inform their course instructor(s) and Accessibility Services of their personal circumstances.

Human Rights, Harassment and Discrimination
The BCIT community is made up of individuals from every ability, background, experience and identity, each contributing uniquely to the richness and diversity of the BCIT community as a whole. In recognition of this, and the intrinsic value of our diversity, BCIT seeks to foster a climate of collaboration, understanding and mutual respect between all members of the community and ensure an inclusive accessible working and learning environment where everyone can succeed. Respect, Diversity, and Inclusion [9] is a supportive resource for both students and employees of BCIT, to foster a respectful learning and working environment. Any student who feels that they are experiencing discrimination or harassment (personal or human rights-related) can confidentially access this resource for advice and support. Please see BCIT Policy 7507 – Harassment and Discrimination [10] and accompanying procedure.

Students should make themselves aware of additional Education, Administration, Safety and other BCIT policies listed at https://www.bcit.ca/about/administration/policies.shtml [11]

Guidelines for School of Computing and Academic Studies

No school specific policies, please refer to main BCIT Policy.

Approved

I verify that the content of this course outline is current.
Medhat Elmasry, Instructor
December 19, 2025

I verify that this course outline has been reviewed.
Chi En Huang, Program Head
January 02, 2026

I verify that this course outline has been reviewed and complies with BCIT policy.
Donna Turner, Associate Dean
January 05, 2026

Note: Students will be given reasonable notice if changes are required to the content of this course outline.

*Course hours and credits are calculated per Policy 5012 [12] and the associated procedure [13] .

Total hours – Example of 3 credit lecture/lab course:

- ***Full-time course****: 45 hours of scheduled learning*
- ***Flexible Learning course****: 36 hours of scheduled learning plus 9 hours of independent (non-scheduled, non-instructional) learning*

List of links found on this page

This list includes all links found on this page for your reference.

[1] https://www.squamish.net/
[2] https://twnation.ca/
[3] https://www.musqueam.bc.ca/
[4] https://www.bcit.ca/files/pdf/policies/5101.pdf
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